

without any need for a centralised server or third party trusted authority.

Till date, Bitcoin is still the most commonly used application using blockchain technology. Bitcoin is a decentralised digital currency payment system that consists of a public transaction ledger called a blockchain.

There are many different kinds of blockchains.

- **Public blockchain:** Public blockchains are like Bitcoins, which are large distributed networks that are run through a native token. They are open for the public to participate at any level and have open source code that the community maintains.
- **Permissioned blockchains:** Permissioned blockchains like Ripple control roles that individuals can play within the network. They are still large and distributed systems that use a native token. Their code may or may not be open source.
- **Private blockchains:** Private blockchains tend to be smaller and don't use a token. Their membership is closely controlled. These types of blockchains are favoured by consortiums that have trusted members and trade in confidential information.

All the above blockchains use cryptography to allow every participant on any given network to manage the ledger in a secure way, without the need for any central authority to enforce the rules.

The underlying principles of the blockchain

The following are the five basic principles underlying the technology.

1. **Distributed database:** Every participant on the blockchain has access to the entire database and its complete history. No single participant controls the data or information and every one of them can verify the records of the transaction partners directly, without any intermediary.
2. **Peer-to-peer transmission:** Communication occurs directly between peers instead of through a central node. Every node stores and forwards information to all the other nodes.
3. **Transparency with pseudonymity:** Every transaction and its associated value is visible to anyone with access to the system. Every node or user on a blockchain has a unique 30+ character alphanumeric address that identifies it. Users can choose to remain anonymous or provide proof of their identity to others. Transactions take place between blockchain addresses.
4. **Record irreversibility:** When the transaction is entered in the database and the accounts are updated, the records cannot be altered because they are linked to every transaction record that is previously recorded. Various types of computational algorithms and proposed approaches are deployed to ensure that the recording on the database is permanent, ordered in proper sequence and available to all others on the network.
5. **Computational logic:** Because of the digital nature of the ledger, blockchain transactions cannot be tied to computational logic

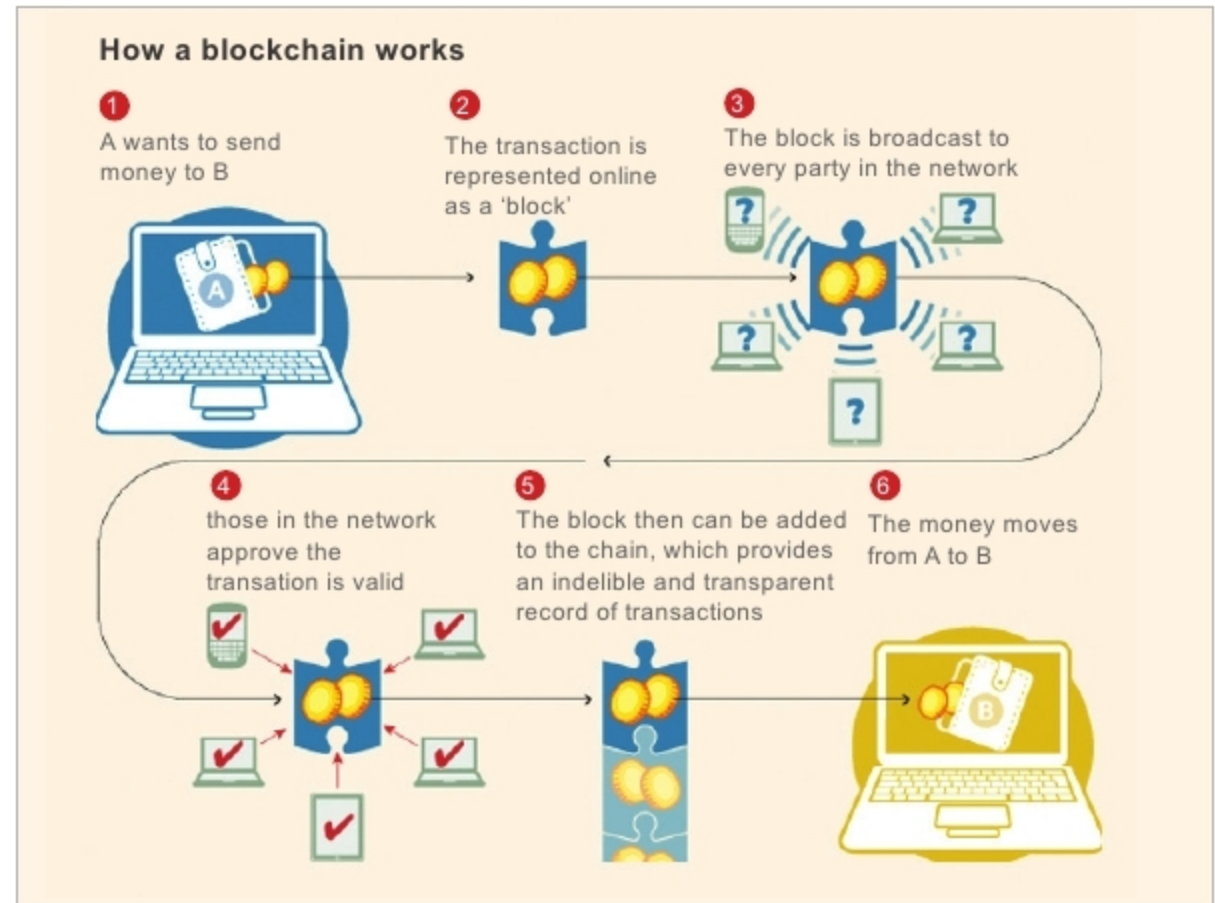


Figure 1: How a blockchain works

and programmed. So, users can set up algorithms and rules that automatically trigger transactions between nodes.

How a blockchain works

The blockchain is a new concept when compared to traditional financial practices and can involve a series of complex steps. The following points highlight the processes in a blockchain. The first few steps are simple to understand, but the subsequent steps become complicated as transactions mature.

1. **Transaction initiation:** In order to run any business, it is necessary to have currency transactions. For example: Party A wants to buy some goods from Party B, for which the former has to perform some cash/online transactions with the latter. Now assume that A and B are Bitcoin users. And A wants to send money to B in exchange for the goods. This is the first stage of the blockchain process.
2. **Identity encryption:** In the blockchain domain, both A and B are in the form of cryptographic keys holding unique public and private keys. The main objective of the private key is to ensure that only that user who owns the key can access